

|                    |                                  |                      |                                   |                                  |
|--------------------|----------------------------------|----------------------|-----------------------------------|----------------------------------|
| <b>TRANSMITTAL</b> | TRANSMITTAL NO.<br>KC-RES-05-T01 | DATE<br>October 2026 | SENT VIA<br>Link · PDF attachment | SHEETS<br>10 sheets + 8 drawings |
|--------------------|----------------------------------|----------------------|-----------------------------------|----------------------------------|

|                                     |                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------|
| TO<br>Hiring team · MEP / BIM group | FROM<br>Kenneth Chow · Burnaby, BC · +1 778-791-3768 ·<br>wmchow5@gmail.com |
|-------------------------------------|-----------------------------------------------------------------------------|

# KENNETH CHOW

**Re:** MEP technician / MEP modeler roles, and coordinator roles where they fit. Enclosed is my BCIT residential MEP model and the code review I ran on it, item by item.

ITEMS ENCLOSED

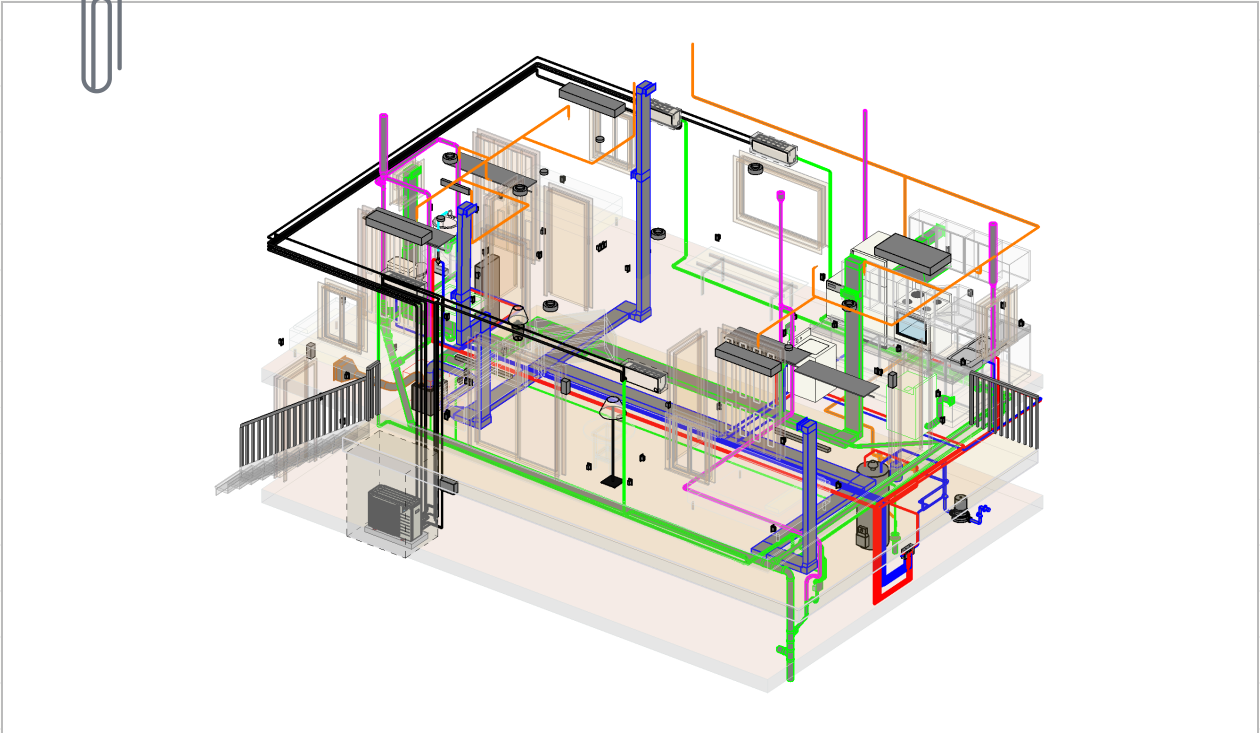
| ENCL. | DESCRIPTION                                                  |
|-------|--------------------------------------------------------------|
| 01    | 3D MEP coordination model, KC-RES-05 (this page)             |
| 02    | Applicant qualifications and model register                  |
| 03    | RFI log: 50 issues raised, 50 closed, 1 reopened on re-check |
| 04    | Six RFIs in full, with code basis and before / after         |
| 05    | Eight drawings exported from the live model                  |

ACTION REQUESTED

- ☒ Review the model and the decisions
- ☒ Invite to interview
- ☐ Revise and resubmit (tell me what to fix)

ENCL. 01 · 3D MEP coordination · view C-3D-OVERVIEW

Exported from Revit 2027, 2 Oct 2026



REMARKS

One-storey house on a basement, Burnaby BC. HVAC, plumbing, fire protection and electrical in one Revit model: 391 pipes (about 451 m), 26 circuits, 28 view templates, 8 sheets.

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| APPLICANT QUALIFICATIONS                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                    | ENCL.<br>02 | APPLICANT<br>Kenneth Chow | BASED IN<br>Burnaby, BC |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------|-------------------------|
| TRAINING<br>Building Information Modeling (BIM) studies, BCIT, 2026.<br>Bachelor of Engineering (Honours), Electronic and Communications, Hong Kong, 2015.                                                                                                                                                                 | MODEL REGISTER · KC-RES-05                                                                                                                                                                                                                                                                         |             |                           |                         |
|                                                                                                                                                                                                                                                                                                                            | + +<br>PIPES 391 · about 451 m                                                                                                                                                                                                                                                                     |             |                           |                         |
| EXPERIENCE<br>Project Coordinator & Electronic Engineer (ELV systems), Shun Hing Engineering Contracting, Hong Kong, Nov 2020 - Sep 2023.<br>Layout drawings, riser diagrams and shop drawings for access control, CCTV and intercom; RFIs, submittals and design markups; wall-cavity checks and power load calculations. | ELECTRICAL CIRCUITS 26 on panel P1 (200 A)                                                                                                                                                                                                                                                         |             |                           |                         |
|                                                                                                                                                                                                                                                                                                                            | VIEW TEMPLATES / FILTERS 28 / 13                                                                                                                                                                                                                                                                   |             |                           |                         |
|                                                                                                                                                                                                                                                                                                                            | SHEETS 8                                                                                                                                                                                                                                                                                           |             |                           |                         |
|                                                                                                                                                                                                                                                                                                                            | ISSUES RAISED / CLOSED 50 / 50                                                                                                                                                                                                                                                                     |             |                           |                         |
|                                                                                                                                                                                                                                                                                                                            | REOPENED ON RE-CHECK 1 (SD1, see R-01)                                                                                                                                                                                                                                                             |             |                           |                         |
| WHAT THAT MEANS HERE<br>I have run the paperwork this page is built from. On KC-RES-05 I applied the same habit to my own model: raise the issue, cite the code, propose options, record the decision.                                                                                                                     | BACKUPS BEFORE EACH CHANGE one per item                                                                                                                                                                                                                                                            |             |                           |                         |
| CODES WORKED TO ON THIS MODEL<br>BC Building Code 2024 Part 9 · BC Plumbing Code · Canadian Electrical Code (as adopted in BC) · NFPA 13D · CSA F280, F326, B52 · CAN/ULC-S553 · Burnaby waterworks bylaw.                                                                                                                 | DESIGN BASIS<br>Equipment selections, fixture locations and sprinkler design flows follow the BCIT project brief. Where the code review called for a change, such as heat-pump indoor units to meet the BCBC heating requirement (M-09), the change and its reasoning are recorded in the RFI log. |             |                           |                         |
| SOFTWARE<br>Autodesk Revit (MEP: piping, ducts, circuits, panel schedules, view templates, families) · AutoCAD · Revit API checks through pyRevit.                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                    |             |                           |                         |
| HOW THE REVIEW WAS DONE<br>The geometry and code checks were scripted through the Revit API with an AI assistant (Claude). I set the rules, chose every option and checked each fix in the model. I can walk through any item on this page.                                                                                |                                                                                                                                                                                                                                                                                                    |             |                           |                         |

| RFI LOG              |       | ENCL.<br>03                                                    | PROJECT<br>KC-RES-05     | RAISED BY<br>K. Chow (self-review)                                 | PERIOD<br>28 Sep - 2 Oct 2026 |
|----------------------|-------|----------------------------------------------------------------|--------------------------|--------------------------------------------------------------------|-------------------------------|
| 51 rows              |       |                                                                |                          |                                                                    |                               |
| RFI                  | DISC. | ISSUE                                                          | CODE BASIS               | RESOLUTION                                                         | STATUS                        |
| C-01                 | HVAC  | No heating: the only outdoor unit was cooling-only             | BCBC 9.33 (22 °C)        | Heat-pump outdoor unit with FTX heads; toe-kick heater in the bath | CLOSED · PLACEHOLDER FAMILY   |
| C-02                 | FP    | Sprinkler SP-07 hidden above the new hall ceiling              | NFPA 13D coverage        | Sidewall head lowered to 115 mm below the ceiling                  | CLOSED                        |
| C-03                 | FP    | Fire main tees straight off domestic water with no components  | NFPA 13D · CSA B64.10    | Control valve, backflow, gauge, flow switch, drain & test          | CLOSED · PLACEHOLDER FAMILY   |
| C-04                 | ARCH  | Deck guards 914 mm and open over a 2.1 m drop                  | BCBC 9.8.8               | 1070 mm guard, balusters 92 mm clear                               | CLOSED                        |
| C-05                 | ARCH  | Exterior stair 610 mm wide, 203 mm run, open risers            | BCBC 9.8                 | 12 × 178 / 11 × 255, 934 mm clear, closed risers                   | CLOSED                        |
| C-06                 | ELEC  | Bathroom switches inside the shower footprint                  | CEC · 500 mm from shower | Light switches on the hall side; fan timer 0.91 m from the shower  | CLOSED                        |
| M-01 <u>FULL RFI</u> | PL    | About 15 drains at 1 % and 4 at 0 %                            | BCPC 2.4.8.1             | All drains ≤ 3" at 2 %; condensate also at 2 %                     | CLOSED                        |
| M-02                 | PL    | Shower had no supply; a tempering valve fed nothing            | CSA B125.3 · 49 °C       | Pressure-balancing shower valve at 1.15 m                          | CLOSED · PLACEHOLDER FAMILY   |
| M-03                 | HVAC  | No range hood; HRV exhaust pickup over the range               | BCBC 9.32 (~50 L/s)      | 150 cfm ducted hood; pickup moved 1.62 m away                      | CLOSED · PLACEHOLDER FAMILY   |
| M-04                 | HVAC  | No dryer exhaust                                               | BCBC 9.32                | 100 mm rigid duct, connected to the dryer, to a wall hood          | CLOSED                        |
| M-05                 | HVAC  | Bathroom exhaust only 40 cfm                                   | BCBC 9.32 (25 L/s)       | 55 cfm through the HRV boost timer                                 | CLOSED                        |
| M-06                 | FA    | No smoke alarm outside Bedroom 3                               | BCBC 9.10.19             | SD6 on the vaulted ceiling, 0.82 m from the peak                   | CLOSED                        |
| M-07                 | FA    | SD1 0.78 m from the AC-01 discharge                            | CAN/ULC-S553             | Moved to 1.70 m from AC-01                                         | CLOSED                        |
| M-08                 | FA    | Smoke alarms reported as not circuited                         | CEC Section 32           | Already on lighting circuits, none on GFCI                         | CLOSED                        |
| M-09 <u>FULL RFI</u> | HVAC  | No F280 calculation; living-room head undersized               | CSA F280                 | FTX15 × 3 + FTX35 heat-pump heads (coursework deviation)           | CLOSED · PLACEHOLDER FAMILY   |
| M-10 <u>FULL RFI</u> | HVAC  | R32 charge against the smallest room                           | CSA B52 / IEC 60335-2-40 | 2.5 kg charge ≤ 2.80 kg allowed                                    | CLOSED                        |
| M-11                 | PL    | Water heater missing T&P valve, expansion tank, straps         | BCPC                     | T&P valve, ¾" copper discharge, 2 gal tank; straps noted           | CLOSED                        |
| M-12 <u>FULL RFI</u> | PL    | PVC pipe running through the water heater tank; no floor drain | BCPC traps               | Floor drain with P-trap (95 mm seal); stack extended               | CLOSED                        |
| M-13                 | ARCH  | East deck stair missing                                        | BCBC 9.8                 | 5 × 173 / 4 × 280, 1.30 m wide, guards both sides                  | CLOSED                        |
| V-01                 | ARCH  | Basement door 1930 mm high                                     | BCBC Table 9.5.5.1       | 813 × 1981 door, 187 mm left for the header                        | CLOSED                        |
| V-02                 | PL    | Water service only 0.26 m below grade                          | Burnaby bylaw 762 mm     | Entry lowered to -1.50 m, about 1.0 m cover                        | CLOSED                        |
| V-03                 | PL    | No cleanout and no sewer connection                            | BCPC                     | Cleanout inside on the stack; 4" sewer at 1 %                      | CLOSED                        |
| V-04                 | ARCH  | Bedroom egress relies on the sprinkler waiver                  | BCBC 9.9.10.1            | Dependency documented on the windows                               | DOCUMENTED                    |
| V-05                 | PL    | Condensate lines 16 mm                                         | Daikin (VP20 / ¾")       | All four lines upsized to ¾"                                       | CLOSED                        |
| P-01                 | HVAC  | AC-01 cuts into the bedroom ceiling                            | Daikin ≥ 50 mm top       | Lowered 130 mm; line sets kept clear of a vent                     | CLOSED                        |
| P-02                 | HVAC  | AC-02 cuts into the ceiling                                    | Daikin ≥ 50 mm top       | Lowered 130 mm                                                     | CLOSED                        |
| P-03                 | HVAC  | AC-03 cuts into the ceiling                                    | Daikin ≥ 50 mm top       | Lowered 130 mm                                                     | CLOSED                        |
| P-04                 | HVAC  | Outdoor unit embedded in the wall, under the deck              | Daikin clearances        | Pulled out 100 mm onto a 100 mm pad                                | CLOSED                        |

| RFI                  | DISC. | ISSUE                                              | CODE BASIS                | RESOLUTION                                                   | STATUS                      |
|----------------------|-------|----------------------------------------------------|---------------------------|--------------------------------------------------------------|-----------------------------|
| P-05                 | ELEC  | Panel P1 runs into the floor joists                | CEC 2-308 working space   | Moved to the west basement wall, 1 m clear                   | CLOSED                      |
| P-06                 | HVAC  | HRV intake labelled return air, exhaust as supply  | System classification     | New Outdoor Air and Exhaust Discharge system types           | CLOSED                      |
| P-07                 | HVAC  | Six 5 cfm token exhaust grilles                    | CSA F326 room rates       | Removed; exhaust moved to bath, kitchen, laundry             | CLOSED                      |
| P-08                 | HVAC  | Supply air into the laundry                        | CSA F326                  | Removed; master bedroom raised to 22 cfm                     | CLOSED                      |
| P-09 <b>FULL RFI</b> | HVAC  | HRV exhaust under the deck, near the outdoor unit  | HRV manual                | HRV rotated and moved; grilles on the west wall 2.47 m apart | CLOSED                      |
| P-10                 | HVAC  | No HRV condensate drain                            | BCPC air gap $\geq 25$ mm | Condensate pump to the receptor, 26 mm air gap               | CLOSED                      |
| P-11                 | PL    | Washer piped rigidly from its base                 | BCPC traps and air break  | Washer box, 2" standpipe and trap                            | CLOSED                      |
| P-12                 | PL    | No main shut-off, check or isolation valves        | BCPC (PRV > 550 kPa)      | Shut-off, check and isolation valves added                   | CLOSED · PLACEHOLDER FAMILY |
| P-13                 | PL    | Dishwasher 3.4 m from the sink                     | BCPC trap and vent        | Kept: coursework location; drain checked compliant           | <b>KEPT BY DESIGN</b>       |
| P-14                 | FP    | Sprinkler flows 32-43 gpm per head                 | NFPA 13D minimums         | Kept: coursework values, conservative                        | <b>KEPT BY DESIGN</b>       |
| P-15                 | FP    | Fire pipe type was the generic "Standard"          | NFPA 13D (copper K)       | Own Copper Type K fire-protection type                       | CLOSED                      |
| P-16                 | FP    | Fire pipes above ceilings may freeze               | NFPA 13D                  | Warm-side installation notes                                 | DOCUMENTED                  |
| P-17                 | FP    | Light L1 50 mm from sprinkler SP-01                | 13D obstruction (24")     | Light moved 0.6 m; 0.65 m clear                              | CLOSED                      |
| P-18                 | ELEC  | Lights floating in vaulted rooms                   | Model accuracy            | Pendant types with stem lengths                              | CLOSED                      |
| P-19                 | ELEC  | Receptacle R25 60 mm from the wall end             | 100 mm from jamb or end   | Moved with switch S9; 115 mm                                 | CLOSED                      |
| P-20                 | ELEC  | Switches hidden in the laundry alcove              | Latch-side placement      | Moved to the alcove wall; hinge side flagged                 | CLOSED                      |
| P-21 <b>FULL RFI</b> | ELEC  | 100 A service with little margin                   | CEC 8-200                 | 200 A service; 62 % with a future EV                         | CLOSED                      |
| P-22                 | ELEC  | No AFCI, weather-resistant or battery-backup specs | CEC 26-658 · BCBC 9.10.19 | Specs on every circuit and device type                       | CLOSED                      |
| P-23                 | BIM   | Confusing level names; elements without a level    | BIM standard              | Levels renamed; 117 schedule levels set                      | CLOSED                      |
| P-24                 | BIM   | Metric family names in an imperial project         | BIM standard              | Renamed to imperial                                          | CLOSED                      |
| P-25                 | BIM   | Window type names didn't match sizes               | BIM standard              | Renamed to real sizes                                        | CLOSED                      |
| P-26                 | BIM   | Architecture and MEP in one file, no links         | BIM practice              | Kept: coursework single file; split plan written             | <b>KEPT BY DESIGN</b>       |
| R-01                 | FA    | SD1 about 0.64 m from supply diffuser AT-06        | CAN/ULC-S553 (~0.9 m)     | Found on re-check; move to about (-3.3, 3.4)                 | <b>OPEN</b>                 |

WHITE - FIRM COPY

# REQUEST FOR INFORMATION

RFI NO.  
C-03

ENCL.  
04-1

DISCIPLINE  
Fire protection

STATUS  
Closed

SUBJECT **Fire sprinkler main fed straight off domestic water**

## QUESTION

The 1¼" sprinkler main teed off the domestic cold water through a transition fitting. No backflow preventer, control valve, gauge or drain & test.

## OPTIONS CONSIDERED

1. Add the standard 13D riser components. 2. Treat it as a multipurpose system. Rejected: the sprinkler piping dead-ends, so water never circulates.

## RESPONSE

Five in-line placeholders on the fire main in the basement, with the drain & test discharging over the floor drain. Then I swapped two of them (see after).

## CODE BASIS

NFPA 13D riser components and waterflow alarm · BC Plumbing Code backflow, CSA B64.10 (type set by the City).

## MY CALL

*Asking what the drain pipe was for exposed the real error: test water left at the drain & test before reaching the flow switch.*

## REFERENCE

FP-CV1 1774562 · FP-BFP1 1774565 · FP-PG1 1774568 · FP-FS1 1774574 · FP-DT1 1774571

## BEFORE

DCW tee → fire main → sprinklers

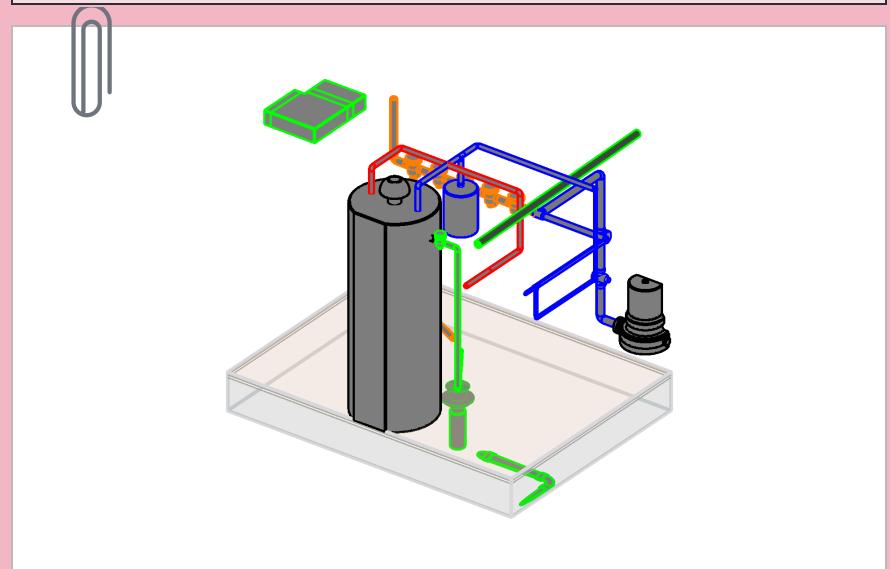
No way to isolate, test, or stop stagnant sprinkler water flowing back into drinking water.

First build order: CV → BFP → PG → D&T → FS. A flow test would never pass the switch.

## AFTER

DCW tee → CV → BFP → PG → FS → D&T → up

Opening the drain & test now sends water through the flow switch first, so the test proves the alarm.



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|                                                                                                                                                                                        |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|
| REQUEST FOR INFORMATION                                                                                                                                                                | RFI NO.<br>M-01 | ENCL.<br>04-2                                                                                                                                                                                                                                                                                                                             | DISCIPLINE<br>Plumbing | STATUS<br>Closed |
| SUBJECT <b>Drainage at 1 % and 0 % slope</b>                                                                                                                                           |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |
| QUESTION<br>About 15 horizontal 1½" and 2" drains were at 1 %, four at 0 %, and the four condensate lines at 1 %.                                                                      |                 | <div>BEFORE</div> <div><p>2" branch <span style="float:right">1 % · 4 at 0 %</span></p><p>Condensate: 16 mm lines at 1 %.</p></div> <div>AFTER</div> <div><p>2" branch <span style="float:right">2 % (¼" per ft)</span></p><p>Exit invert unchanged at the time; 4" building drain stays at 1 %. Clash check clean.</p></div> <div></div> |                        |                  |
| OPTIONS CONSIDERED<br>Re-slope sanitary only and leave condensate at 1 % (common HVAC practice), or follow the code for both.                                                          |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |
| RESPONSE<br>Every drain 3" and under now at 2 %, upstream ends raised so the 4" building drain and its exit stay put. Condensate at 2 % with five new corner drops to clear crossings. |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |
| CODE BASIS<br>BC Plumbing Code 2.4.8.1: drains ≤ 3" at least 1 in 50 (2 %); over 3" may be 1 in 100. Condensate to a receptor is indirect waste, which is drainage pipe.               |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |
| MY CALL<br><i>I asked: "under BC law, is 1 % fine? If not, follow the law."</i>                                                                                                        |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |
| REFERENCE<br>Kitchen and DW main · KS-01 arm · washer branch · shower · condensate IW1-IW4                                                                                             |                 |                                                                                                                                                                                                                                                                                                                                           |                        |                  |

# REQUEST FOR INFORMATION

RFI NO.  
M-09 · M-10

ENCL.  
04-3

DISCIPLINE  
HVAC

STATUS  
Closed

SUBJECT **Heat-pump sizing and the R32 charge limit**

## QUESTION

No CSA F280 calculation. The living room's 2.0 kW head served a 35 m<sup>2</sup> room with a 5.9 m vault. The heads were cooling-only, so the house had no real heat.

## OPTIONS CONSIDERED

A. Heat-pump FTX heads on a heat-pump outdoor unit. B. Keep the coursework cooling-only heads and heat with about 2.3 kW of extra baseboard, which pushes the panel past 100 A.

## RESPONSE

Option A, recorded as a deviation from the BCIT equipment spec (FTC15NV14 × 3 + FTC20NV14). Then checked the R32 charge against the smallest room.

## CODE BASIS

BCBC 9.33 heating to CSA F280 · CSA B52 / IEC 60335-2-40 charge limit for A2L refrigerant.

## MY CALL

*"The AC units are a coursework requirement... but I know option A is more efficient. Go ahead anyway."*

## REFERENCE

AC-01 1524859 · AC-02 1525077 · AC-03 1525454 · AC-04 1526631 · CU-01 1527078

## BEFORE

AC-04 FTC20: **2.0 kW** against an open-zone cooling load of **3.3 kW**.

All four heads FTC: cooling only. Heating came from 500 W baseboards.

## AFTER

F280-STYLE ESTIMATE house 3.6 kW heat / 4.6 kW cool

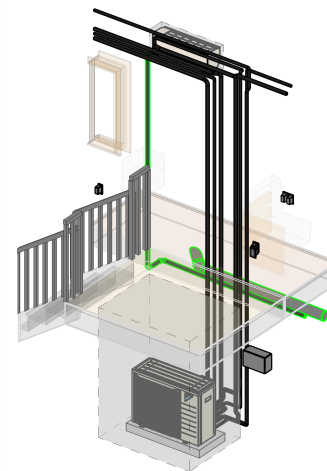
HEADS FTX15 × 3 + FTX35

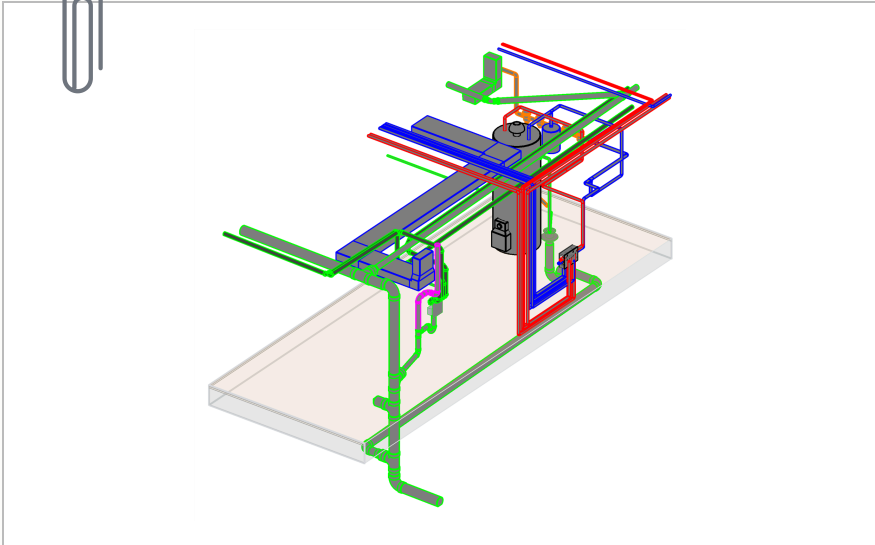
LIQUID LINES 18.9 + 5.8 + 8.9 + 22.3 = 55.9 m

CHARGE 2.0 + (55.9 - 30) × 0.02 = **2.5 kg**

m<sub>max</sub> (Bedroom 1) 2.5 × 0.307<sup>1.25</sup> × 1.8 × √7.44 = **2.80 kg**

RESULT 2.5 ≤ 2.80, passes with no mitigation



| REQUEST FOR INFORMATION                                                                                                                                                                                                         |  | RFI NO.<br>M-12 | ENCL.<br>04-4                                                                                                                                                                                                                         | DISCIPLINE<br>Plumbing | STATUS<br>Closed |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|
| SUBJECT <b>A pipe through the water heater, and no floor drain</b>                                                                                                                                                              |  |                 |                                                                                                                                                                                                                                       |                        |                  |
| <div>QUESTION</div> <div>A 1" PVC pipe hung off the water heater's drain-valve connector with its centreline inside the tank wall. Its comment said it was the T&amp;P discharge. The basement had no floor drain for it.</div> |  |                 | <div>BEFORE</div> <div>1" PVC on the tank's <b>drain valve</b>, clipping the tank. PVC is not rated for 99 °C relief water.</div> <div>No floor drain; building drain exit only 0.14 m below the floor.</div>                         |                        |                  |
| <div>OPTIONS CONSIDERED</div> <div>(a) Extend the building drain stack down and drain a floor drain by gravity. (b) A sump and pump.</div>                                                                                      |  |                 | <div>AFTER</div> <div><div>WH-01</div> → <div>TPV-01</div> → <div>¾" Cu</div> → <div>air gap</div> → <div>FD-01</div> → <div>P-trap</div> → <div>stack</div></div> <div>Trap seal about 95 mm; trap seal primer noted on FD-01.</div> |                        |                  |
| <div>RESPONSE</div> <div>Deleted the pipe; added a real T&amp;P valve with a ¾" copper discharge; floor drain FD-01 with a P-trap, 3" drain at 2 % to a new sanitary tee; stack extended from -2.19 to -2.95 m.</div>           |  |                 | <div></div>                                                                                                                                       |                        |                  |
| <div>CODE BASIS</div> <div>BC Plumbing Code: T&amp;P discharge full size and temperature-rated, no valves; every fixture trapped, seal ≥ 38 mm; floor drains need trap seal protection.</div>                                   |  |                 |                                                                                                                                                                                                                                       |                        |                  |
| <div>MY CALL</div> <div><i>I rejected two trap placements, then placed the trap myself the right way up and had the method written into a reusable skill.</i></div>                                                             |  |                 |                                                                                                                                                                                                                                       |                        |                  |
| <div>REFERENCE</div> <div>FD-01 1782449 · trap 1783819 · tee 1782637 · TPV-01 1782320 · WH-01 1602886</div>                                                                                                                     |  |                 |                                                                                                                                                                                                                                       |                        |                  |

# REQUEST FOR INFORMATION

RFI NO.  
P-09

ENCL.  
04-5

DISCIPLINE  
HVAC

STATUS  
Closed

SUBJECT **HRV exhaust discharging under the deck**

## QUESTION

The HRV exhaust grille sat under the deck, 0.53 m from the outdoor heat pump's clearance zone.

## OPTIONS CONSIDERED

Keep it · slide it east on the south wall · run a long duct to the west wall · rotate the HRV · rotate and move the HRV. Compared in one table.

## RESPONSE

Rotated HRV-01 90° and moved it beside the west wall: both grilles on the west wall, 2.47 m apart, out from under the deck. Then added the receptacle and condensate pump it was missing.

## CODE BASIS

HRV manufacturer separation ( $\geq 1.8$  m) and height above grade ( $\geq 450$  mm); keep moist exhaust off deck framing and the heat-pump coil.

## MY CALL

*The discharge duct looked too long, so I suggested rotating the HRV, then asked about its power socket.*

## REFERENCE

HRV-01 1547124 · AT-10 1567942 · AT-17 1785468 · REC-HRV 1785491 · CP-01 1788255

## BEFORE

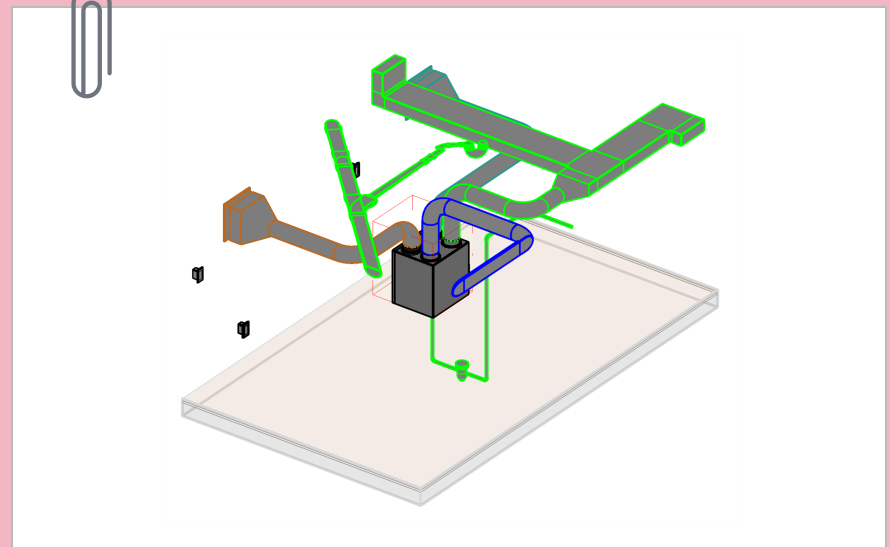
Exhaust under the deck, **0.53 m** from the heat-pump clearance zone. Moist air on deck joists, frost on the coil.

Outdoor ducting about 4.5 m. No receptacle and no condensate drain at the HRV.

## AFTER

Grilles on the west wall, **2.47 m** apart; discharge about 3.5 m from the heat pump.

Outdoor ducting about 3.0 m straight. Dedicated receptacle REC-HRV; condensate pump CP-01 to the receptor with a 26 mm air gap.



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|                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|------------------|
| <h1>REQUEST FOR INFORMATION</h1>                                                                                                                                                                              | RFI NO.<br>P-21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ENCL.<br>04-6 | DISCIPLINE<br>Electrical | STATUS<br>Closed |
| SUBJECT <b>Service size and a future EV charger</b>                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |
| QUESTION<br>The 100 A service had little room for heating, a heat pump or an EV charger.                                                                                                                      | <div> <div> <b>BEFORE</b><br/>           Area    <math>89.6 + 75 \% \times 89.6 = 156.8 \text{ m}^2</math><br/>           Basic    <math>5\,000 + 1\,000 = 6\,000 \text{ W}</math><br/>           Heat pump + HTR-4    <math>= 4\,000 \text{ W}</math><br/>           Range (<math>\leq 12 \text{ kW}</math>)    <math>= 6\,000 \text{ W}</math><br/>           Water heater    <math>= 4\,500 \text{ W}</math><br/>           25 % of dryer, pump    <math>= 1\,750 \text{ W}</math><br/> <div> <math>22\,250 \text{ W} \div 240 \text{ V} = </math> <b><u>93 A on 100 A</u></b> </div> </div> <div> <b>AFTER</b><br/> <math>22\,250 \text{ W} + \text{EV } 7\,680 \text{ W} = 29\,930 \text{ W} \div 240 \text{ V} = 125 \text{ A}</math><br/>           Service now 200 A → <b><u>62 %</u></b> (target <math>\leq 80 \%</math>)<br/>           Slots 30/32 reserved: 40 A 2-pole EV spare, no outlet yet         </div> </div> |               |                          |                  |
| OPTIONS CONSIDERED<br>A. Keep 100 A with an EV energy-management system. B. Upgrade to 200 A.                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |
| RESPONSE<br>Ran the CEC 8-200 calculation from the model's loads. 93 A on 100 A complies. I set my own target of $\leq 80 \%$ including a future EV, so P1 went to 200 A with a reserved 40 A two-pole spare. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |
| CODE BASIS<br>CEC 8-200 single-dwelling calculation: service rating $\geq$ calculated load. The 80 % rule (8-104) is for continuous loads, not this calculation.                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |
| MY CALL<br><i>"93 % of the load would offend any laws?" No. Then: "I would like to make it 80 % including future EV charging."</i>                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |
| REFERENCE<br>Panel P1 1785080 · spare slots 30/32                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                          |                  |

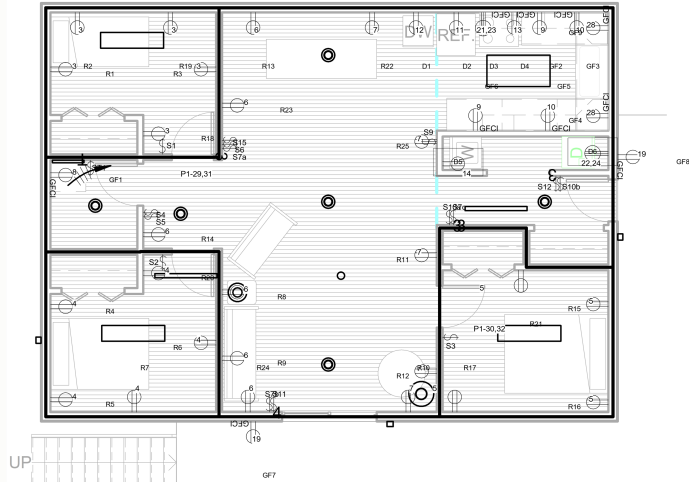
# DRAWING ENCLOSURES

ENCL.  
05 A-H

SOURCE  
Revit views, exported 2  
Oct 2026

NOTE  
Title blocks cropped

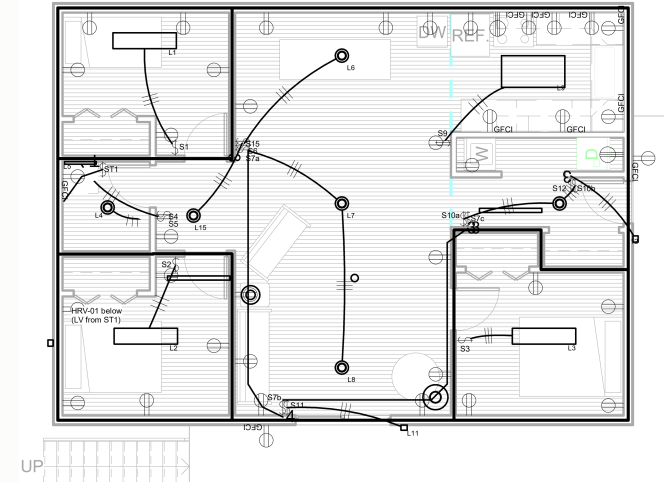
ENCL. 05-A · Power plan, Level 1



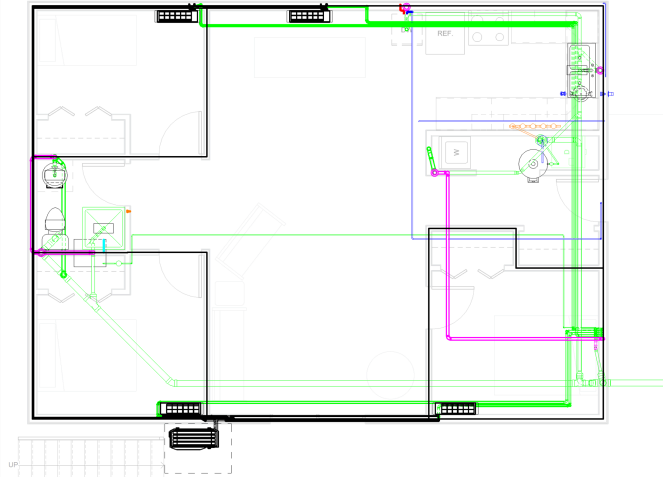
ENCL. 05-A · POWER PLAN, LEVEL 1

[Click the drawing to enlarge](#)

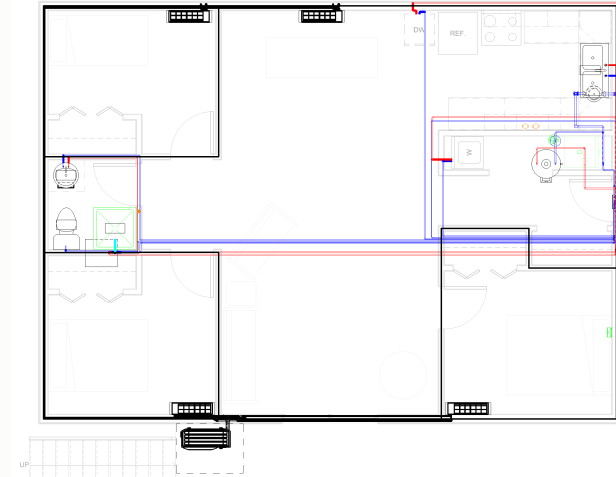
ENCL. 05-B · LIGHTING PLAN, LEVEL 1

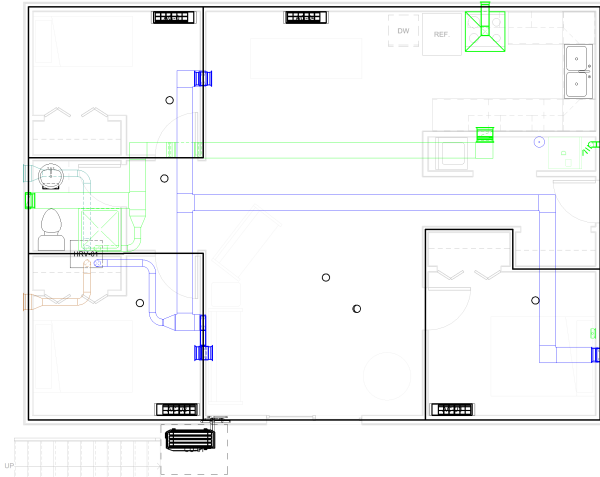


ENCL. 05-C · SANITARY DRAINAGE AND VENT

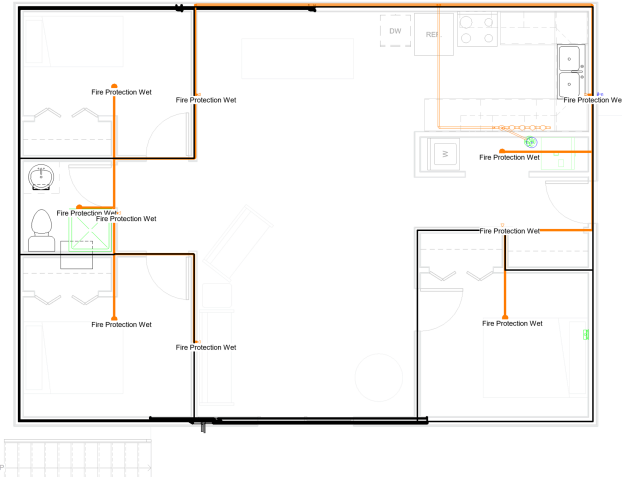


ENCL. 05-D · DOMESTIC WATER

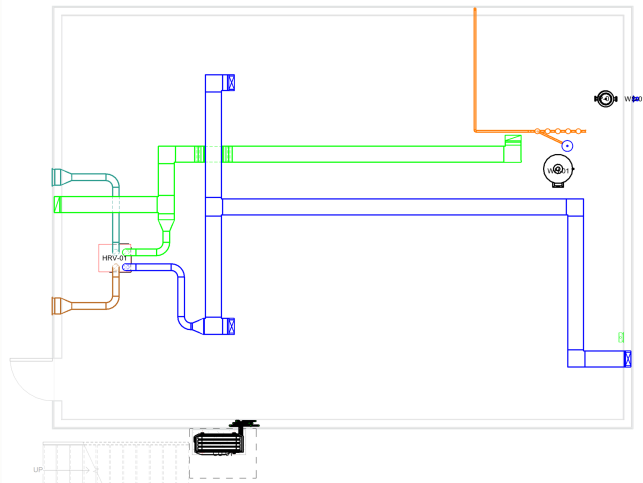




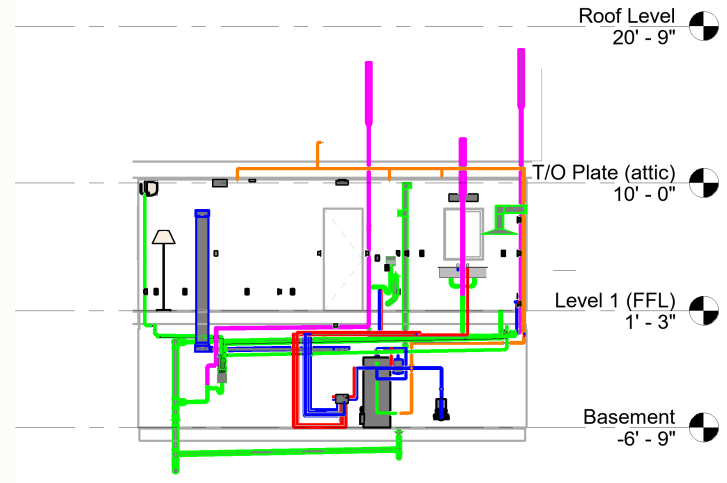
ENCL. 05-E · HVAC, LEVEL 1



ENCL. 05-F · FIRE SPRINKLERS, LEVEL 1



ENCL. 05-G · BASEMENT PLANT



ENCL. 05-H · COORDINATION SECTION 01

## REVIEWER ACTION

TRANSMITTAL NO.  
KC-RES-05-T01

RETURN TO  
Kenneth Chow

EMAIL **wmchow5@gmail.com**

PHONE **+1 778-791-3768**

BASED IN **Burnaby, BC**

### OPEN ITEMS CARRIED FORWARD

R-01: smoke alarm SD1 sits about 0.64 m from supply diffuser AT-06; it needs about 0.9 m. Found on my own re-check after the log closed. Fix: move SD1 to about (-3.3, 3.4).

Placeholder families (heat pump, valves, shower valve, range hood) stand in for manufacturer families. They cause the 2 remaining Revit warnings.

### PREPARED BY

Kenneth Chow · BIM & design coordination · BCIT BIM studies 2026.  
Model review assisted by Claude through the Revit API; every decision is mine and I can explain it.

~~CANARY - RETURN COPY~~